

Assignemnet – 1

$k = 2$, $k = 3$ and $k = 4$

Id	X	Y
1	1	2
2	2	3
3	3	4
4	4	5
5	5	6
6	6	7
7	7	8
8	8	9
9	9	10
10	10	11
11	11	12
12	12	13

$k = 2$ divide into two groups

Calculate the cenroids

Calculate the distance with out square root

$$(x_2 - x_1)^2 + (y_2 - y_1)^2$$

Group1: (1,2), (2,3) (3,4) (4,5) (5,6) (6,7) : $\frac{1+2+3+4+5+6}{6}$, $\frac{2+3+4+5+6+7}{6} = \frac{21}{6}, \frac{27}{6} =$
(3.5,4.5)

$$(1,2) \text{ vs } (3.5,4.5): (1 - 3.5)^2 + (2 - 4.5)^2$$

$$(2,3) \text{ vs } (3.5,4.5): (2 - 3.5)^2 + (3 - 4.5)^2$$

$$(3,4) \text{ vs } (3.5,4.5): (3 - 3.5)^2 + (4 - 4.5)^2$$

$$(4,5) \text{ vs } (3.5,4.5): (4 - 3.5)^2 + (5 - 4.5)^2$$

$$(5,6) \text{ vs } (3.5,4.5): (5 - 3.5)^2 + (6 - 4.5)^2$$

$$(6,7) \text{ vs } (3.5,4.5): (6 - 3.5)^2 + (7 - 4.5)^2$$

Group2: (7,8), (8,9) (9,10) (10,11) (11,12) (12,13)

k = 3 divide into two groups

Calculate the centroids

Calculate the distance with out square root

$$(x_2 - x_1)^2 + (y_2 - y_1)^2$$

k = 4 divide into two groups

Calculate the centroids

Calculate the distance with out square root

$$(x_2 - x_1)^2 + (y_2 - y_1)^2$$

WCSS: Within cluster sum of squares

Is used to identify the number of clusters

As we know data can be divided into many clusters

k = 1 to n

k = 1 means all the data points consider as one cluster

k = n means there are 'n' clusters = number of data points

Goal:

To choose best clusters between 1 to n

- 1) Within cluster sum of squares (WCSS)*
- 2) Silhouette score*
- 3) Hierarchical cluster (one cluster algorithm)*

Within cluster sum of squares (WCSS)

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

Step – 1: Randomly divide the groups based k value

$K = 2$ (Two groups)

Group – 1 : (1,2,3)observations

(1,1) (1.5,2) (3,4)

Group – 2 : (4,5,6,7)observations

(5,7) (3.5,5) (4.5,5), (3.5,4.5)

$$\text{Mean}(G_1) = \frac{(1 + 1.5 + 3)}{3}, \frac{1 + 2 + 4}{3} = (1.83, 2.33)$$

$$\text{Mean}(G_2) = \frac{(5 + 3.5 + 4.5 + 3.5)}{4}, \frac{7 + 5 + 5 + 4.5}{4} = (4.125, 5.5)$$

$$(1,1) \text{ vs } (1.83, 2.33): (1.833 - 1)^2 + (2.33 - 1)^2 = d_{11}$$

$$(1.5, 2) \text{ vs } (1.83, 2.33): (1.833 - 1.5)^2 + (2.33 - 2)^2 = d_{21}$$

$$(3, 4) \text{ vs } (1.83, 2.33): (1.833 - 3)^2 + (2.33 - 4)^2 = d_{31}$$

$$\text{cluster} - 1: d_{11} + d_{21} + d_{31}$$

$$(5,7) \text{ vs } (4.125,5.5): (4.125 - 5)^2 + (5.5 - 7)^2 = d_{42}$$

$$(3.5,5) \text{ vs } (4.125,5.5): (4.125 - 3.5)^2 + (5.5 - 5)^2 = d_{52}$$

$$(4.5,5) \text{ vs } (4.125,5.5): (4.125 - 4.5)^2 + (5.5 - 5)^2 = d_{62}$$

$$(3.5,4.5) \text{ vs } (4.125,5.5): (4.125 - 4.5)^2 + (5.5 - 5)^2 = d_{72}$$

$$cluster - 2: d_{42} + d_{52} + d_{62} + d_{72}$$

$WCSS = \text{sum of } c_1 \text{ distance} + \text{sum } c_2 \text{ distances}$

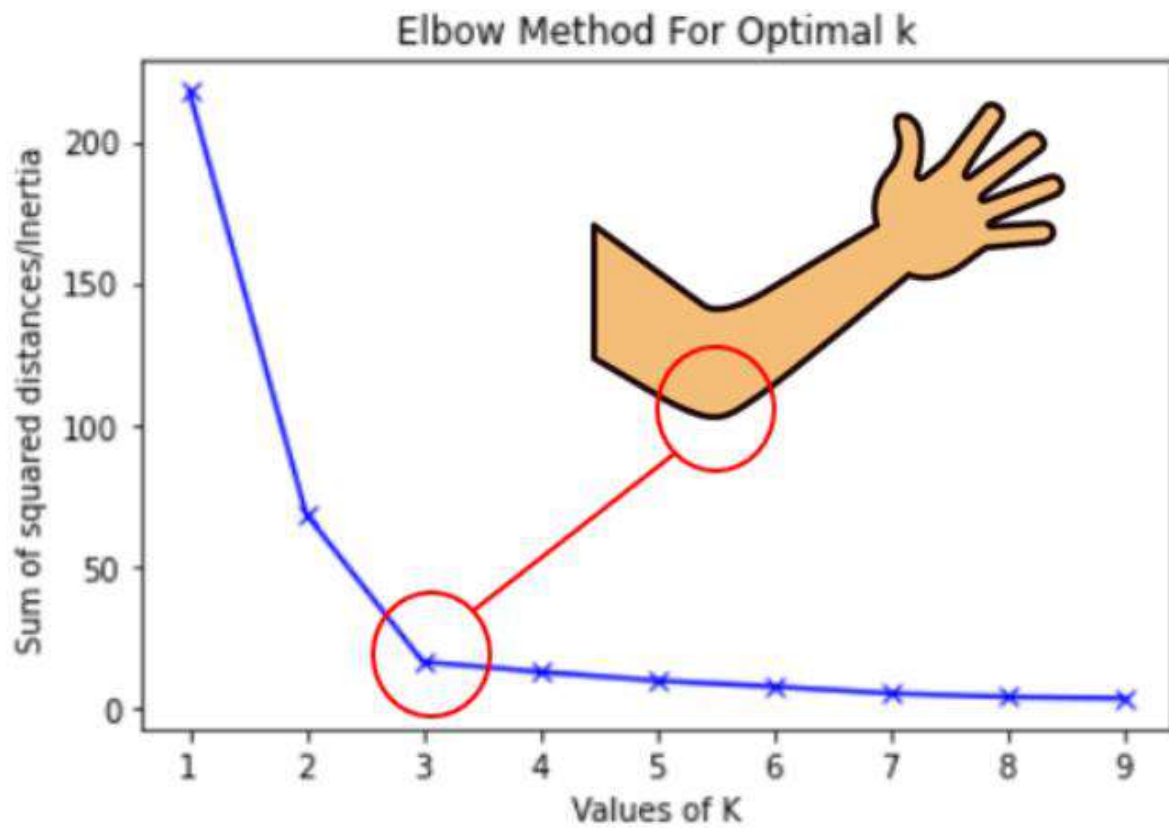
For $k = 2$ will get WCSS this also called as Inertia

How the data points very close to the centroids

Cluster	WCSS
K=1	2000
K=2	1500
K=3	1000
K=4	800
K=5	700
K=6	650

Graph between these two Cluster vs wcss

Elbow method



Line plot between K and inertia

For $k = 2$

Total data points are 12 divided into two clusters

First 6 points are one clusters , second 6 points are another cluster

$$d_{1c1} + d_{2c1} + d_{3c1} + d_{4c1} + d_{5c1} + d_{6c1} + d_{7c2} + d_{8c2} + d_{9c2} + d_{10c2} + d_{11c2} + d_{12c2}$$

d_{31} = distance of 3rd point with first cluster centroid

d_{102} = distance between 10th data point with second cluster centroid

Cluster	WCSS
K=2	70
K=3	30
K=4	16

silhouette score

SS will give the idea , the corresponding data point belongs to suitable cluster or not

It gives an indea about How a datapoint is well clustered or not

Supppose we divided data points into two parts

Step – 1: Randomly divide the groups based k value

$K = 2$ (Two groups)

Group – 1 : (1,2,3)observations

(1,1) (1.5,2) (3,4)

Group – 2 : (4,5,6,7)observations

(5,7) (3.5,5) (4.5,5), (3.5,4.5)

**Each data point will calculate the distance between other data point in withinthe cluster
as well as data points with other clusters**

x_1, x_2, x_3 are belongs to cluster – 1

x_4, x_5, x_6, x_7 are belongs to cluster – 2

consider x_1 point calculate the distance with all the other data points in cluster – 1

$d(x_1, x_2)$ $d(x_1, x_3)$

Now take the average of distance =

$$\frac{[d(x_1, x_2) + d(x_1, x_3)]}{2} = a$$

x_1 point calculate the distance with all the other data points in cluster – 2

$d(x_1, x_4)$ $d(x_1, x_5)$, $d(x_1, x_6)$ $d(x_1, x_7)$

Now take the average of distance =

$$\frac{[d(x_1, x_4) + d(x_1, x_5) + d(x_1, x_6) + d(x_1, x_7)]}{4} = b$$

$$S = \frac{b - a}{\max(b, a)}$$

Case – 1:

assume that x_1 point belongs to cluster – 1

what could be the s value?

In this scenario b value high compare to a : $b \gg \gg \gg a$

$$s = \frac{(b - a)}{b}$$

$$b - a \cong b$$

$$s = 1$$

$s = 1$ means the corresponding data point very well clustered

Case – 2:

assume that x_1 point on the border of the two clusters

what could be the s value?

In this scenario b equal to a : $b = a$

$$s = 0$$

$s = 0$ means the corresponding data point on the border

Case – 3:

assume that x_1 point belongs to cluster – 2 but we place in cluster – 1

x_1 to cluster₁: a

x_1 to cluster₂: b

$$S = \frac{b - a}{\max(b, a)} = -\frac{a}{a} = -1$$

S varies from -1 to 1

-1 means wrongly grouped

0 on the border

1 Means correctly grouped